TOKEN LIKELIHOOD CAN LIE: AUDITING PHYSICAL ALIASING IN TOKENIZED WORLD-MODEL PLANNING

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ABSTRACT

Tokenized world models compress continuous observations or states into discrete codebook tokens and predict future token sequences. We audit a selection-time failure: a planner samples N token futures and executes the one with the highest internal token score. In controlled tokenized/VQ settings, larger N improves selected token likelihood while worsening real decoded utility when the codebook aliases physically distinct futures. We call this token-likelihood physical aliasing. We give an exact finite, tie-aware law for the selected real-utility curve under top-score sampling and validate it by Monte Carlo with mean absolute error 0.00276. In the main controlled run, raw selection at $N = 64$ raises selected token likelihood to 1.386 while real utility falls to 0.0556 and alias/physical-violation rates reach 1.0. A combined token-specific repair raises real utility to 0.629 and drops both rates to 0.0, while an oracle reaches 0.957, leaving a clear gap. The results are diagnostic rather than universal: they do not establish external benchmark performance, hardware deployment evidence, or a general indictment of tokenized world models.

1 INTRODUCTION

Discrete world tokens are appealing because they turn continuous, high-dimensional prediction into a compact symbolic or codebook-indexed sequence problem. A tokenizer can absorb visual detail, a learned transition model can predict future token sequences, and a planner can search over token futures rather than raw states. This pattern appears in VQ-style representation learning, latent world models, and model-based planning systems that separate a compact predictive state from downstream selection.

The same bottleneck creates a planning-specific risk. A token future can be likely under the token model while representing several physically different decoded futures. If the internal score sees only the token sequence, decoded reward, or a token critic, it can assign high score to a future whose real execution is low value because hidden modes, physical validity, or decoder consistency were collapsed by the codebook. We study this as *token-likelihood physical aliasing*.

Score-tail selection stresses the issue. A planner samples N candidate futures and selects the highest-scoring one. Increasing N concentrates selection on the upper tail of the score distribution. When the score tail is aligned with real utility, this is useful. When the score tail is physically aliased, larger N can select invalid decoded futures more reliably. The failure is therefore not "more samples are bad"; it is "more samples search harder through whatever score tail the token model exposes."

This paper is a diagnosis-and-repair package for that mechanism. The contribution is deliberately scoped:

1. We formalize finite token-future selection and give an exact tie-aware top-score law for the expected selected real utility of any fixed empirical candidate pool.
2. We identify token-likelihood physical aliasing as a concrete failure mode in which token likelihood rises while decoded/executed utility falls.
3. We provide controlled VQ/tokenized evidence, including learned codebook artifacts, codebook-size sweeps, decode-validity diagnostics, and exact-law validation.

4. We test token-specific repairs: codebook uncertainty, alias detection, decode-consistency penalties, physical-validity filtering, rare-mode reweighting, and a small token-to-real pilot calibrator.
5. We package the evidence with an audit discipline that limits claims to controlled settings where aliasing is measurable or pilot labels are collected.

Our claim is not that tokenized world models are broadly flawed, nor that score-tail selection is broadly unsafe. The claim is narrower and testable: if the selected score aliases physical utility, high-N selection can amplify the aliasing tail; token-specific diagnostics and pilot calibration can repair this in controlled detectable cases.

2 SETUP

Let $x_{0:H}$ denote a physical future and $z_{0:H} = q_\phi(x_{0:H})$ its tokenized representation under a learned codebook. A tokenized world model samples future token sequences from a generator $p_\theta(z_{1:H} \mid z_0, a_{0:H})$ or from an induced candidate pool. A decoder or downstream evaluator maps tokens back to predicted observations, rewards, or constraints.

For each candidate future c_i, we distinguish three quantities:

- a token score S_i, such as token likelihood, decoded reward, or a learned token critic;
- a real utility R_i, measured after decoding or executing the corresponding physical future with hidden-mode and validity penalties;
- diagnostics D_i, including alias probability, quantization error, decode-consistency error, physical-violation indicators, and rare-token statistics.

The separation between S_i and R_i is essential. In the controlled environment used here, the tokenizer is trained on observable coordinates while a hidden physical mode influences real utility. This makes it possible for multiple physical futures to share token cells or near-identical token scores while differing in validity and utility. We call a candidate *aliased* when its token representation hides a physically important distinction that affects R_i but is weakly represented in S_i.

Score-tail selection draws N candidates independently from a finite empirical pool and returns an index with maximal score, breaking score ties uniformly. The selected real utility is R_N^{sel}. We study the curve

$$N \mapsto \mathbb{E}[R_N^{\mathrm{sel}}],$$

and compare it for raw token scores, repaired scores, and an oracle real-utility score.

3 EXACT FINITE LAW

The central law is finite-population, tie-aware, and score-agnostic. It applies to token likelihood, a repaired diagnostic score, an oracle score, or any other scalar selector.

Theorem 1 (Tie-aware finite score-tail law)**.** *Consider a finite candidate pool $\{(S_i, R_i)\}_{i=1}^m$. Partition the candidates into score-tie groups g sorted by increasing score. Let L_g be the number of candidates with score strictly lower than group g, let U_g be the number of candidates with score no higher than group g, and let $\bar{R}_g$ be the mean real utility within group g. If N candidates are sampled iid uniformly from the pool and the highest-score tie group is broken uniformly, then*

$$\mathbb{E}[R_N^{\mathrm{sel}}] = \sum_g \bar{R}_g \left[\left(\frac{U_g}{m}\right)^N - \left(\frac{L_g}{m}\right)^N \right]. \tag{1}$$

The proof is a one-line decomposition over the event that the maximum sampled score falls in group g; Appendix A gives the details. Equation 1 immediately explains both success and failure. The expected selected score is nondecreasing in N because the sample maximum is nondecreasing. The expected selected real utility has no such monotonicity unless high-score groups also have high real utility. A low-utility top score group becomes more likely as N grows.

The law is not asymptotic and does not assume continuous scores. This matters for tokenized world models because ties and near-ties are common: codebook cells merge many continuous states, token likelihoods can be quantized or smoothed, and decoded reward can saturate.

4 Token-Specific Diagnostics and Repairs

The repair methods target the token bottleneck rather than generic score smoothing. Each candidate receives the raw token score

$$S_i^{\mathrm{raw}} = 0.70 \cdot \ell_i^{\mathrm{tok}} + 0.35 \cdot \hat{r}_i^{\mathrm{tok}} - 0.04 \cdot e_i^{\mathrm{q}} + \epsilon_i,$$

where ℓ_i^{tok} is token likelihood, $\hat{r}_i^{\mathrm{tok}}$ is decoded reward, e_i^{q} is quantization error, and ϵ_i is small score noise in the controlled generator. Real utility is evaluated separately and penalizes hidden-mode mismatch and physical invalidity.

We evaluate six repair families.

Codebook uncertainty. This score penalizes token-level alias probability and normalized quantization error. It asks whether the selected candidate lies in a codebook cell known to merge hidden modes or reconstruct poorly.

Decode consistency. This score penalizes disagreement between token-implied futures and decoded physical futures, together with physical-violation scores. It is designed for cases where the token model predicts a plausible code sequence that decodes into an impossible trajectory.

Physical filter. This score applies a hard penalty when a candidate crosses a physical-violation threshold, then applies a lighter alias penalty to the remaining candidates. In this paper the filter uses generated diagnostics; in a deployment it would require task-specific validity checks.

Rare-mode reweighting. This score rewards rare tokens while penalizing alias and violation diagnostics. It tests whether the raw score is discarding rare but valid high-utility futures.

Token-to-real pilot calibration. A small ridge calibrator maps token diagnostics to real utility using a pilot fraction of labeled candidates. This is the only repair that uses real-utility labels. It is therefore powerful but not free.

Combined repair and gate. The combined score averages codebook uncertainty, decode consistency, physical filtering, and pilot calibration with fixed weights. A deployment gate inspects high-N alias rate, physical-violation rate, raw utility drop, repair gain, and exact-law error. In the main run the gate returns `collect_pilot_labels`, because pilot calibration repairs a harmful high-N tail but the raw selector is unsafe.

5 Experiments

All experiments are CPU-local controlled experiments. The goal is to isolate the token-tail mechanism, not to claim external benchmark or hardware performance. The repository includes every primary artifact under `results/` and `figures/`.

World-token construction. We generate synthetic physical observations with an omitted hidden mode, train a small VQ tokenizer, and estimate token frequency, transition statistics, quantization error, and alias probability. The main learned artifact uses codebook size 8, collision rate 0.232, mean quantization error 0.120, token entropy 2.946, and transition self-probability 0.137.

Candidate pools. For each of 150 pools with 64 candidates, the generator emits a mixture of valid candidates, aliased candidates, rare-good candidates, and low-quality candidates. The raw score favors high token likelihood and decoded reward. Real utility is clipped to $[-0.05, 1.0]$ after hidden-mode and physical-validity penalties. We evaluate $N \in \{1, 2, 4, 8, 16, 32, 64\}$.

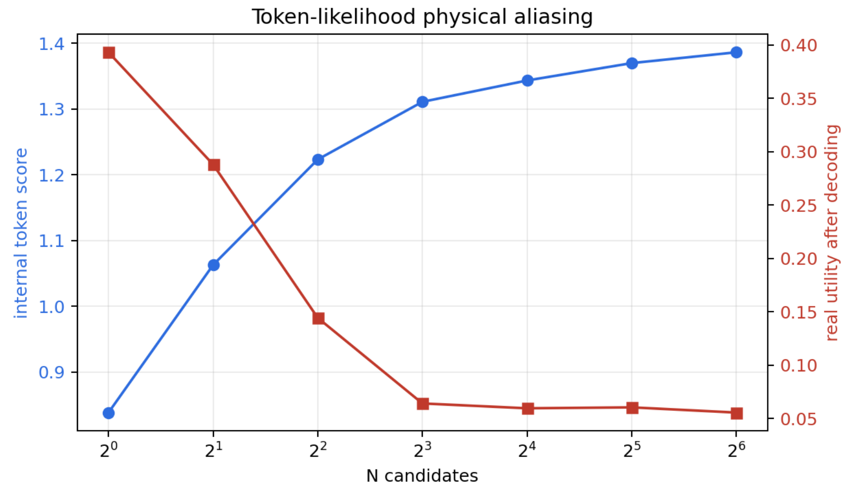

Figure 1: Raw score-tail selection amplifies token-likelihood physical aliasing. As N grows, selected token likelihood rises, but selected real utility falls because the high-score tail is dominated by aliased and physically invalid decoded futures.

Table 1: Main selected-tail statistics. Raw high-N selection improves token likelihood while collapsing real utility. Repairs reduce detected aliasing and physical invalidity, but oracle selection remains substantially better.

Selector	N	Token likelihood	Real utility	Alias rate	Violation rate	Decode error
Raw token score	1	0.838	0.393	0.387	0.467	0.450
Raw token score	64	1.386	0.0556	1.000	1.000	0.613
Combined repair	64	0.755	0.629	0.000	0.000	0.274
Pilot calibrator	64	0.463	0.819	0.000	0.000	0.299
Oracle real utility	64	0.488	0.957	0.000	0.000	0.317

Artifacts. The main metrics are in `results/metrics_main.csv`; tail autopsy rows are in `results/tail_autopsy.csv`; exact-law validation is in `results/exact_law_validation.json`; and claim status is in `results/claims_status.json`. Figure 1 shows the raw failure curve.

5.1 Raw High-N Selection Selects the Wrong Tail

Raw token selection looks better if one watches only token likelihood. At $N = 1$, selected token likelihood is 0.838 and selected real utility is 0.393. At $N = 64$, selected token likelihood rises to 1.386, but real utility falls to 0.0556. Alias rate and physical-violation rate both reach 1.0, and decode-consistency error reaches 0.613.

This is exactly the tail effect predicted by Equation 1. The selected score improves because the maximum score improves. The selected real utility falls because the upper score group is dominated by aliased candidates.

5.2 Repair Helps, But Does Not Close the Oracle Gap

Figure 2 compares raw selection, individual repairs, combined repair, the pilot calibrator, and oracle selection. The combined repair raises real utility at $N = 64$ from 0.0556 to 0.629 and drops both alias and violation rates from 1.0 to 0.0. The standalone pilot calibrator reaches 0.819, showing the value of small real-utility labels in this controlled setting. Oracle real-utility selection reaches 0.957.

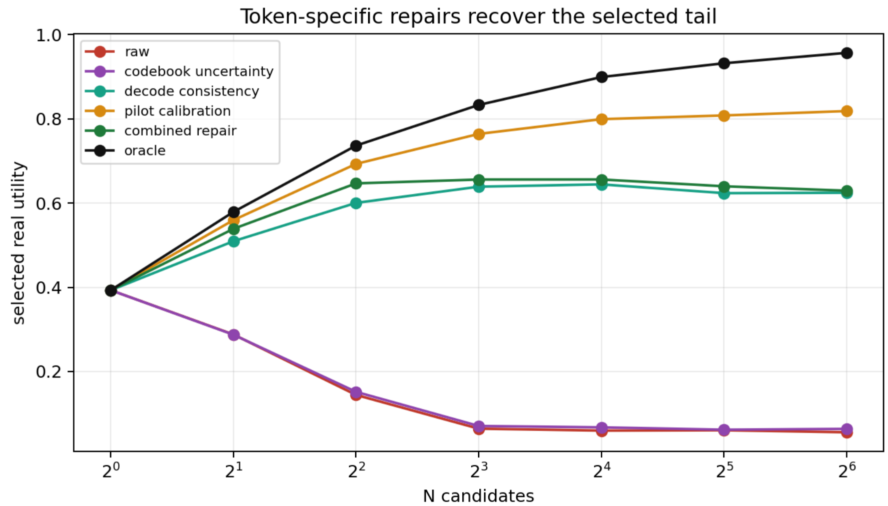

Figure 2: Token-specific repairs can redirect high-N selection away from aliased futures. The strongest label-based pilot calibrator is intentionally separated from hand-coded diagnostics because it spends real-utility labels.

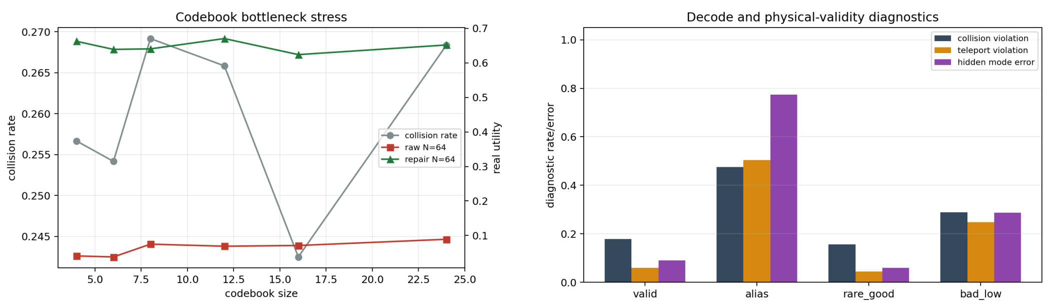

Figure 3: Left: increasing codebook size reduces quantization error but does not remove hidden-mode collisions in this controlled setup. Right: token-specific diagnostics separate raw high-score aliased candidates from repaired valid selections.

The gap is important. The combined repair leaves an oracle gap of 0.327 at $N = 64$, and even the pilot calibrator leaves a gap of 0.138. We therefore treat repair as conditional evidence for detectable aliasing, not as a universal fix.

5.3 Codebook Bottlenecks and Decode Diagnostics

The codebook-size sweep tests whether the failure is merely a small-codebook artifact. As codebook size increases from 4 to 24, mean quantization error decreases from 0.159 to 0.070, but collision rates remain near 0.24–0.27 because the hidden physical mode remains omitted. Raw $N = 64$ real utility stays low, between 0.038 and 0.089, while combined repair gains remain between 0.553 and 0.622.

Decode diagnostics localize the selected bad tail. In the tail autopsy, raw $N = 64$ selections are aliased candidates with high token likelihood, high physical-violation scores, and low real utility. Combined repair selections are mostly valid candidates with lower violation scores. Figure 3 shows both the codebook sweep and the diagnostic distributions.

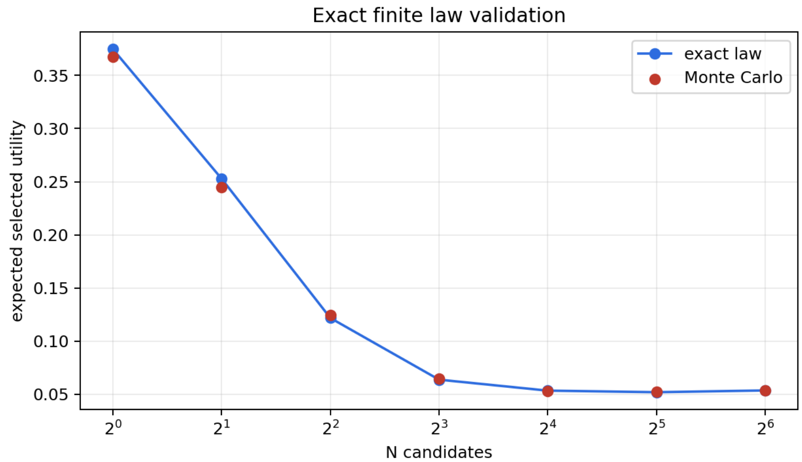

Figure 4: The finite tie-aware law predicts the selected real-utility curve for the fixed empirical token-future pool. The residual is Monte Carlo estimation error, not an asymptotic approximation gap.

5.4 Exact-Law Validation

For fixed empirical pools, Equation 1 gives the expected selected real utility exactly. We validate the implementation against Monte Carlo draws over $N \in \{1, 2, 4, 8, 16, 32, 64\}$. Mean absolute error is 0.00276 and maximum error is 0.00779, consistent with simulation noise. Figure 4 plots exact and Monte Carlo curves.

6 Related Work

Discrete and tokenized representations. Vector-quantized autoencoders introduced learned discrete latent codebooks for neural representation learning (van den Oord et al., 2017). Later work used discrete latents for high-fidelity image and video generation (Razavi et al., 2019; Esser et al., 2021). Our focus is not generation quality; it is the planning consequence of selecting from token-future score tails when a codebook cell hides physically relevant distinctions.

World models and model-based planning. World models learn predictive latent dynamics for decision making (Ha & Schmidhuber, 2018; Hafner et al., 2019; 2020). Model-based RL and MPC systems use learned dynamics to plan by sampling or optimizing candidate futures (Chua et al., 2018; Nagabandi et al., 2018; Williams et al., 2017). We study a narrow inference-time selection law inside such systems: the generator can be held fixed while the selected score tail changes with N.

Reranking, verifiers, and score-tail selection. Sampling many candidates and selecting with a verifier, reward model, or critic is common in sequence generation and decision systems (Cobbe et al., 2021; Nakano et al., 2021; Stiennon et al., 2020). The same pattern appears in planning as trajectory sampling plus scoring. Our contribution is to make the selected real-utility curve finite and auditable for tokenized world futures, especially under score-utility aliasing.

Calibration and safety filtering. Calibration methods relate model scores to empirical outcomes (Zadrozny & Elkan, 2002; Guo et al., 2017). Selective prediction and safety filters abstain or constrain decisions when confidence or validity checks fail (Geifman & El-Yaniv, 2017; Wabersich & Zeilinger, 2021). Our deployment gate is in this spirit, but specialized to token-future diagnostics: alias rate, decode consistency, physical validity, repair gain, and pilot token-to-real labels.

7 Limitations and Claim Discipline

The experiments are synthetic and controlled. The tokenizer, hidden mode, candidate generator, and validity penalties are designed to expose the mechanism. This is useful for diagnosis, but it is not an external benchmark result and not hardware deployment evidence. The learned VQ artifact is small, and the candidate pools are generated from a controlled mixture rather than a large-scale embodied world model.

The repairs are conditional. Codebook uncertainty helps only when aliasing is reflected in codebook statistics. Decode consistency helps only when invalid decoded futures are detectable. Physical filtering requires a meaningful validity check. Pilot calibration requires representative real-utility labels and can drift if the deployment distribution changes. The deployment gate therefore returns `collect_pilot_labels` in the main run rather than declaring raw high-N selection safe.

The finite law is exact for a fixed empirical pool and iid sampling with uniform tie breaking. It does not by itself solve distribution shift, adaptive candidate generation, correlated rollouts, or misspecified real utility. Those cases require fresh audited pools or stronger assumptions.

8 Conclusion

score-tail planning over world tokens is a tail-selection problem. Increasing N reliably searches the upper tail of the token score, but whether that improves real utility depends on whether the score tail is physically meaningful. In our controlled tokenized/VQ setting, raw high-N selection amplifies token-likelihood physical aliasing: token likelihood rises, real utility falls, and selected futures become aliased and physically invalid. The exact finite law predicts this curve, and token-specific diagnostics plus pilot calibration repair the detectable case while leaving an honest oracle gap. The practical message is simple: before increasing N for tokenized world-model planning, audit the token-score tail against real utility or use a gate that asks for pilot labels, repair, or abstention.

Reproducibility Statement

The repository includes source code, tests, generated CSV/JSON artifacts, figures, and this LaTeX source. The main reproduction commands are `bash scripts/run_all.sh`, `bash scripts/run_claim_audit.sh`, `pytest`, and `powershell -ExecutionPolicy Bypass -File paper/iclr/build.ps1`. Appendix G maps the paper build and audit commands.

Ethics and Impact Statement

This work uses synthetic CPU-local experiments and does not use human subjects, private data, deployed robots, or safety-critical hardware trials. The main risk is over-interpreting controlled diagnostic evidence as deployment certification. The paper explicitly limits the claim and recommends task-specific audits before using high-N selection in new systems.

LLM Usage Statement

Large language models were used for drafting and packaging assistance. The authors are responsible for the claims, code, citations, and final text. Numerical claims are taken from tracked repository artifacts.

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A Proof of the Finite Law

Let the finite pool contain m candidates and let group g contain all candidates with one tied score value. The event that the selected maximum score lies in group g is the event that all N iid draws have score no higher than group g and not all draws have score strictly lower than group g. Since

U_g candidates have score no higher than the group and L_g candidates have score strictly lower, the probability of this event is

$$\left(\frac{U_g}{m}\right)^N - \left(\frac{L_g}{m}\right)^N.$$

Conditional on this event, uniform tie breaking within the selected score group gives expected utility $\bar{R}_g$. Summing these disjoint events over score groups proves Equation 1. Replacing $\bar{R}_g$ with the group mean score yields the expected selected-score curve.

B Experiment Details

The controlled generator is implemented in `src/token_alias_tail_audit/simulation.py`. It first builds synthetic observations with an omitted hidden physical mode, fits a small VQ codebook, and estimates per-token frequency, alias probability, quantization error, and transition statistics. Candidate pools are then sampled from four candidate kinds:

- `alias`: high token likelihood and decoded reward, high physical violation, hidden-mode error, and low real utility;
- `rare_good`: lower token likelihood, low physical violation, and high real utility;
- `bad_low`: low score and low utility;
- `valid`: moderate score, low violation, and moderate utility.

The main run uses 150 pools, pool size 64, and $N \in \{1, 2, 4, 8, 16, 32, 64\}$. All methods are evaluated on the same generated candidates. The exact law is validated against Monte Carlo draws from the raw-score empirical pool.

Table 2: Learned VQ artifact used in the main run.

Quantity	Value
Codebook size	8
Collision rate	0.232
Mean quantization error	0.120
Token entropy	2.946
Mean transition self-probability	0.137

C Repair Score Definitions

Let S^{raw}, a, q, d, v, and b denote raw score, alias probability, quantization error, decode-consistency error, physical violation, and rare-token bonus. The implemented diagnostic scores are

$$S^{\text{codebook}} = S^{\text{raw}} - 0.85a - 0.25q,$$
$$S^{\text{decode}} = S^{\text{raw}} - 0.95d - 0.75v,$$
$$S^{\text{filter}} = \begin{cases} S^{\text{raw}} - 2.0, & v > 0.45, \\ S^{\text{raw}} - 0.35a, & v \le 0.45, \end{cases}$$
$$S^{\text{rare}} = S^{\text{raw}} - 0.75a + 0.35b - 0.55v.$$

The pilot calibrator is a ridge regressor from eight token diagnostics to real utility, trained on the first 16% of generated candidates with a minimum of twelve pilot labels. The combined repair is

$$S^{\text{combined}} = 0.15S^{\text{codebook}} + 0.15S^{\text{decode}} + 0.25S^{\text{filter}} + 0.45S^{\text{calibrated}}.$$

These weights are fixed in the repository implementation and are not tuned per figure.

D Additional Diagnostics

Figure 5 visualizes the alias atlas. The purpose is diagnostic rather than aesthetic: token cells with higher alias probability and larger decode/physical errors are the cells that raw high-N selection tends to over-select.

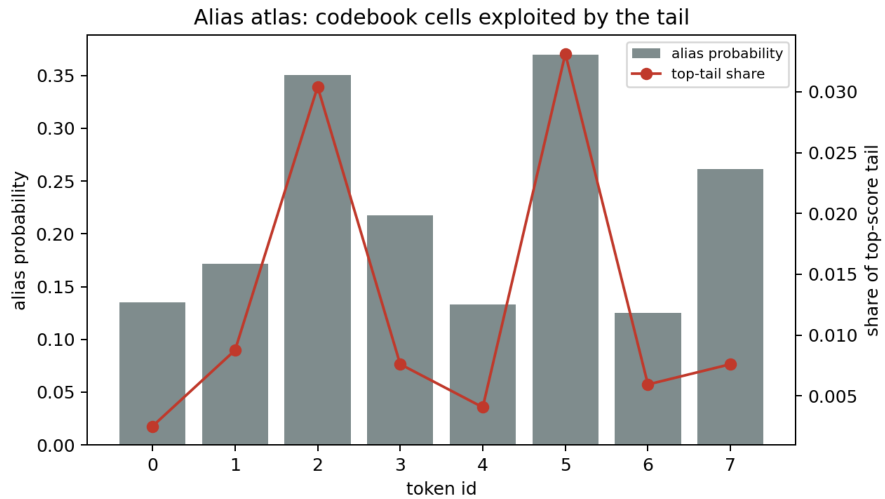

Figure 5: Alias atlas for the controlled tokenized world model. Raw high-score selection concentrates in token regions where hidden-mode aliasing and decoded physical invalidity are high.

E Artifact Map

Table 3: Primary artifact map.

Claim family	Evidence
Raw aliasing failure	`results/metrics_main.csv`; `figures/figure1_token_likelihood_physical_aliasing.png`
Repair comparison	`results/metrics_main.csv`; `figures/figure2_repair_comparison.png`
Codebook bottleneck	`results/codebook_bottleneck.csv`; `figures/figure3_codebook_bottleneck_curve.png`
Decode diagnostics	`results/tail_autopsy.csv`; `figures/figure4_decode_validity_diagnostics.png`; `figures/figure6_alias_atlas.png`
Exact finite law	`results/exact_law_validation.json`; `figures/figure5_exact_law_validation.png`; `src/token_alias_tail_audit/law.py`
Claim audit	`results/claims_status.json`; `results/claims_status.md`; `docs/final_audit.md`

F Claim Checklist

G Reproduction Commands

From the repository root:

```
bash scripts/run_all.sh
bash scripts/run_claim_audit.sh
pytest
powershell -ExecutionPolicy Bypass -File paper/iclr/build.ps1
```

The paper source is in `paper/iclr/`. The build script writes `paper/iclr/main.pdf` and copies the submission PDF to `C:/Users/wangz/Downloads/tokenized_world_model_iclr2027_submission.pdf`.

Table 4: Submission claim checklist.

Check	Status
Scientific object is a tokenized/VQ world model.	Yes
Generator emits discrete world-token futures.	Yes
Scorer includes token likelihood and decoded reward.	Yes
Real utility is measured separately from token score.	Yes
Central failure is token-likelihood physical aliasing.	Yes
Learned/semi-learned VQ artifact is included.	Yes
Repair methods are token-specific.	Yes
Exact finite law is implemented and validated.	Yes
External benchmark or hardware deployment evidence is claimed.	No
Universal repair or universal high-N harm is claimed.	No